

were showcased at Bharat Mobility Expo 2025, the West Asia crisis has hastened the shift as policymakers look to ensure energy security amid elevated crude oil prices and rising import bill. To kickstart the rollout of FFVs which can run on up to 100% ethanol, the ministry of road transport and highways in April included E100 (pure ethanol) in automobile testing and certification standards.

“Any product that could take anything above 85% ethanol could not have been classified as an FFV. That has been removed. It opens the possibility to E100 also,” says Vikram Gulati, Country Head & EVP, Corporate Affairs & Governance, Toyota Kirloskar Motor.

Even though the E20 fuel mandate has been a rushed affair, FFVs have industry backing. And the blueprint already exists—and it comes from Brazil, one of the world’s earliest and most successful adopters of FFVs.

With the introduction of FFVs, the government may not have to increase the base ethanol blending rate, says Gulati. “In Brazil, while the base blend remains E27, the average national blend has risen to as much as 50% with FFVs,” he says. FFVs account for around 90% of new car sales in Brazil, the world’s second-largest producer of ethanol after the United States.

“In Brazil, the ethanol offtake after the introduction of FFVs was a J curve, as enablers were put in place. Consumers found economic rationale to use ethanol instead of their base blend,” says Gulati. With FFVs, concerns around corrosion and mileage loss get resolved, he says. “It’s a no-brainer solution for our country. It’s the best for all stakeholders: consumers, oil marketing companies, government, ethanol producers and automakers.” Flex-fuel engines

can run on 100% ethanol. They automatically detect ethanol content and adjust performance.

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Though it makes little sense for consumers to buy FFVs when fuel is not available, two Japanese automakers have introduced flex-fuel motorcycles in India. In 2024, Honda Motorcycle & Scooter India launched the CB300F Flex Fuel motorbike that supports E85. Suzuki launched the Gixxer SF 250 at the Bharat Mobility Expo in 2025. TVS Motor Company and Bajaj Auto have also showcased flex-fuel motorcycle prototypes. Royal Enfield, the maker of Bullet and Classic motorbikes, is also readying its portfolio for higher ethanol blends.

“We have been exporting to Brazil for over 10 years,” says B Govindarajan, Managing Director of Eicher Motors Ltd and CEO of Royal Enfield. “We have one ready product, the Classic 350, with flex fuel compatibility. We are now looking at the rest of our product line. The moment the infrastructure is ready for FFVs, we will be ready with the products,” says Govindarajan.

Hero MotoCorp, the country’s largest two-wheeler manufacturer, launched its maiden flex-fuel vehicles on June 3—Hero HF Deluxe Flex Fuel and Hero Splendor Plus Flex Fuel. “As soon as flex fuel starts becoming available, the industry can introduce FFVs pretty much in the same month,” Harshavardhan Chitale, CEO of Hero MotoCorp, told *Business Today* in an interview. “Many OEMs are ready with the products. One can expect announcements in the near future,” he says. Flex-fuel variants of Hero HF and Splendor have been approved by the authorities, says Chitale.

FLEX-FUEL VEHICLES ON THE ANVIL

- Maruti Suzuki Fronx FFV
- Maruti Suzuki WagonR FFV
- Tata Punch FFV
- Toyota Innova Hycross FFV
- Hyundai Creta FFV

NUMBERS AT A GLANCE

E85

A fuel blend containing 85% ethanol and 15% petrol.

E100

Pure ethanol

| 95

90%

the penetration of flex-fuel vehicles in Brazil

FFVs to cost around

₹50,000

more than a conventional ICE car

Ethanol has

30%

less energy density than petrol

